

Demonstrating Freeform Fabrication of Fluidic Edible Materials

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ABSTRACT

From providing nutrition to being social platforms, food plays an essential role in our daily lives and cultures. In this work, we are interested in using food as an interaction medium and a context of personal fabrication with an expanded design space enabled by support bath-assisted printing. The bath scaffolds the embedded materials while preserving shapes during the printing processes and allows us to create freeform food with fluid-like materials. Coupled with different post-processing and cooking methods, this technique grants the versatility of food printing with fluidic materials. We will demo confectionery arts and dishes designed by our software tool.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; **Interaction devices**; **Haptic devices**;

KEYWORDS

Food printing, edible interface, personal fabrication, computational fabrication, 3D printing

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1 INTRODUCTION

Beyond providing nutrition, food and dining also play an irreplaceable role across cultures and human activities. Recently, the HCI community has also become interested in food as an interaction platform, including using food to mediate user behaviors and sensory experiences [2, 7, 8], display information [5, 6, 10], create shape-changing interfaces [10, 11], and enable personalization [9, 13],

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most of which use subtractive (e.g. CNC milling) and extrusion-based additive methods to customize shapes. Yet, additively extruding a material, i.e., the ‘ink’, requires it to be at an exact viscosity level that has to be fine-tuned with extrusion nozzle diameter, flow rate, temperature, etc. The objects created are also solid and have minimal overhang to avoid collapses, narrowing down the design space.

To expand the shape space and materials library available for chefs, hobbyist makers, and home cooks, we adopt a bioprinting technique, embedded printing [4], which uses hydrogel baths as temporary scaffolds and applies it to various food ingredients. This modification extends the design space of food printing by including materials that are fluidic and/or require long curing time and replicating shapes that are inherently 3D and freeform with high fidelity. Embedded printing, since its invention, has been modified to suit different engineering [1, 12] and design [3] purposes. To our best knowledge, this work is the first to apply embedded printing to the fabrication of food.

2 METHOD OVERVIEW

Having perused a rich literature on how scientists prepare support baths for embedded printing, we compare and contrast two types of support baths to accommodate different food ink properties – gelatin, an animal tissue derivative, and Carbopol, a polyacrylic acid powder which is food safe. To prepare the support bath with either medium, gelatin, sold as dry sheet or powder, is made into a block of gel and blended into fine particles to form a slurry, and Carbopol powder is stirred in alkaline water to form a 1% solution that feels like hair gel. Both support baths provide omnidirectional floatation to the extruded ink, and when the extrusion nozzle travels at a speed, the support bath is locally sheared and thinned to allow the ink to come out from the nozzle.

When choosing the right support bath, the food ink’s pH value and printing temperature are two important factors as the gelatin and Carbopol support baths react to both factors differently. We derive the guidelines as shown in Figure 1 to help users understand the compatibility of ink and support bath and choose the best fabrication pipeline that works for a particular ink. All of our demo examples follow these guidelines as color-coded.

Similar to support bath selection, the print-release methods (i.e., ways of taking a finished print out of the support bath except

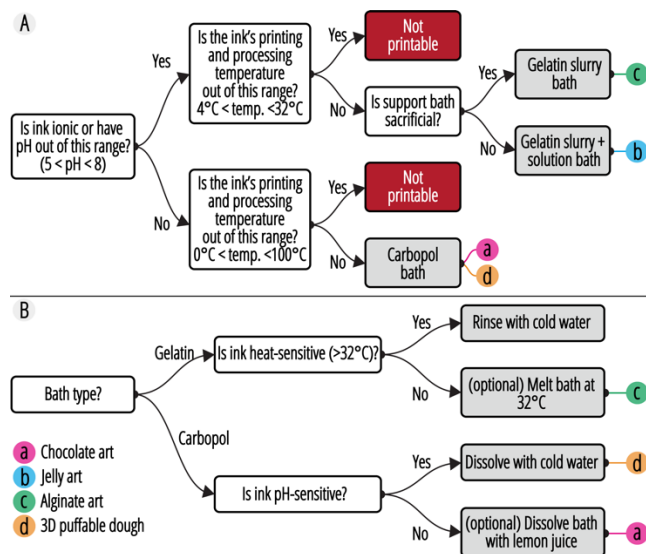


Figure 1: Guidelines for (A) selecting support bath types and (B) release methods.

making the support bath part of the display) also depend on the material properties. Specifically, the gelatin bath can be dissolved by heat, and the Carbopol bath can be liquified with acidic juice. If these methods are incompatible with the printed material, users are advised to use cold distilled water to rinse away the supports.

We implement a suite of design tools in Rhinoceros 3D and Grasshopper to allow users to convert geometries into freeform fabrication files. In addition to the layer-by-layer and parallel line infills common in conventional 3D printing, we implement several new functions to accommodate the generation of non-planar toolpaths. A freeform slicer can take a set of 3D surfaces as input and produce toolpaths by finding their intersections (Figure 2A). Non-planar layers or surfaces can be rasterized with geodesically offset lines to create dense infills (Figure 2B) with equidistant spacing. Lastly, users can also assign toolpaths with non-uniform material flow rates to create variable diameters (Figure 2C). Finally, this suite of design tools will generate a .gcode file to be fed into a 3D printer.

3 APPLICATION EXAMPLES

We use the following examples to showcase the printing fidelity, freeform design, and interactivity of food made of fluidic materials, all of which were printed using a consumer-grade printer retrofitted with a syringe pump or a commercial modular printer. All our materials are purchased from food suppliers. Specifically, gelatin, calcium chloride (CaCl_2), and alginate are purchased from Modernist Pantry. Vegetables, chocolate, syrup, flour, and oil are purchased from a local grocery store.

Freeform chocolate printing: using the Carbopol support bath and our design tools suite allows us to create freeform structures (Figure 4C) with ease. The ink used in this example is an off-the-shelf chocolate bar, thus resembling the texture similar to conventional chocolate treats with high printing fidelity.

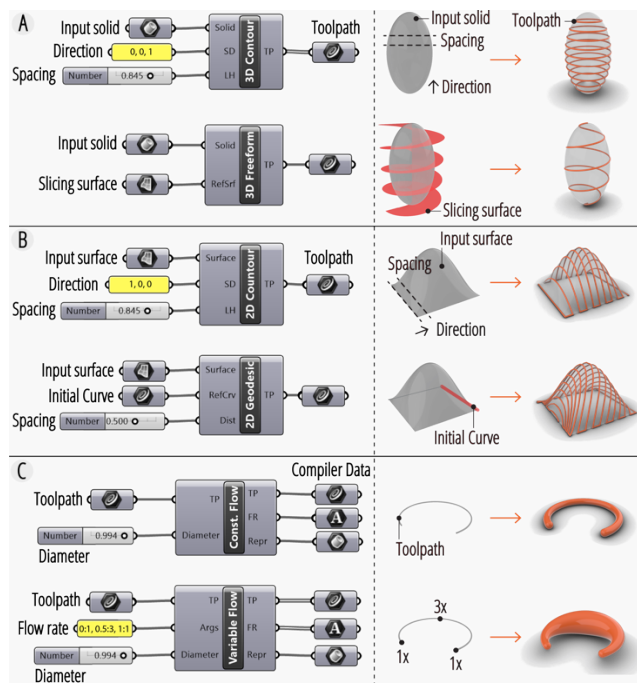


Figure 2: Toolpath design functions - (A) solid to toolpath and (B) surface to toolpath slicing, and (C) toolpath flow rate assignment.

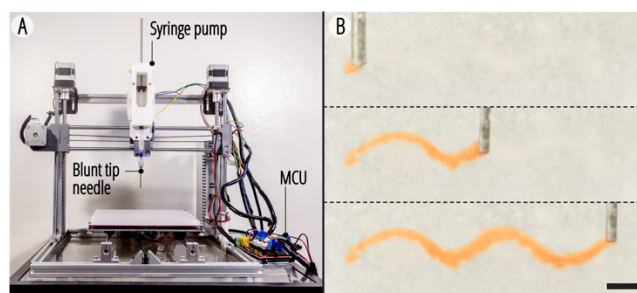


Figure 3: Fabrication system - (A) printer setup and (B) printing process (gelatin bath with alginate-carrot ink). Scale bar: 5 mm.

Jelly art: ‘wagashi’ (Japanese-style confection) uses gelatin as one of the main ingredients, and their making process involves the finesse and craftsmanship of the chef. In this design example, our technique allows hobbyist makers to design and make personalized dishes in their kitchens. Figure 5 shows several wagashi arts designed and made by the authors – who have no prior experience in making wagashi – in their own houses. In addition to the jelly printed with our blended carrot ink, Figure 5B also shows a design made of chocolate syrup and printed with a variable printing diameter inside a container.

Solid-to-liquid bouillon: in addition to the jelly art that traps the embedded ink inside, we show that the gelatin slurry bath can also be used as a sacrificial bath that constitutes the dish itself. Figure 6

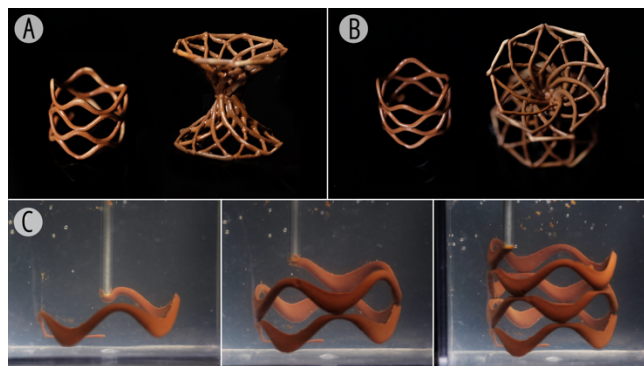


Figure 4: Freeform chocolate art - (A) side and (B) top view, and (C) printing process.

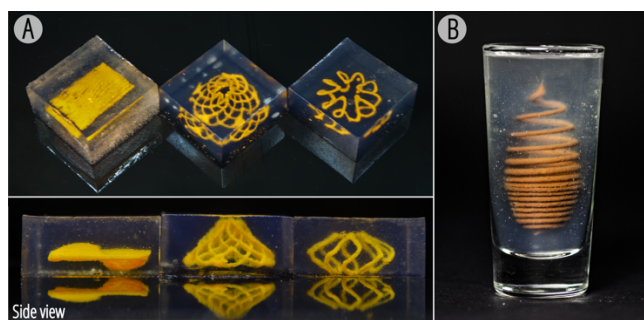


Figure 5: Jelly arts (wagashi) printed with (A) alginate-carrot and (B) chocolate syrup ink.

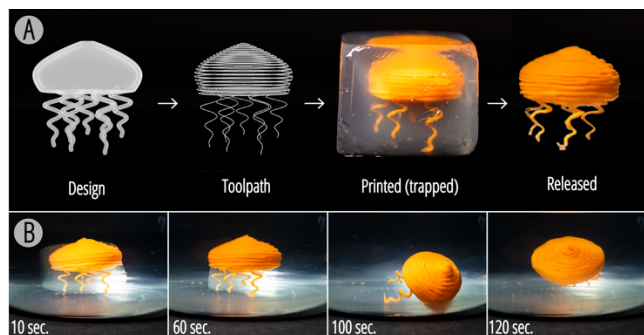


Figure 6: A jellyfish made of alginate-carrot ink - (A) target shape, printing toolpath, and printed and released jellyfish. (B) The release process showing the gelatin dissolving away in a water bath (60°C) and the jellyfish becoming adrift.

shows the design, fabrication, and release process of the alginate jellyfish. The jellyfish and the support bath slurry were removed from the printing container as a whole and submerged into a hot water bath above 60°C to dissolve away the slurry, becoming a soup dish. Taking this idea further, we believe that similar entrap-and-release designs can also be used for targeted flavor and nutrient release.

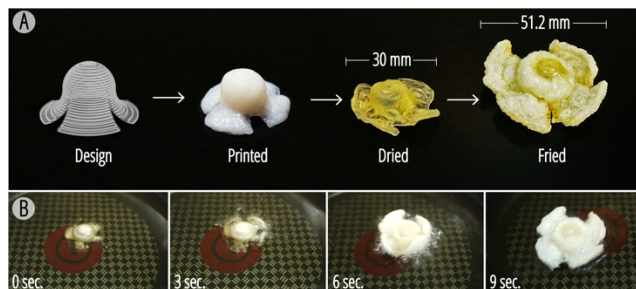


Figure 7: 3D puffable dough - (A) digital models and fabricated objects, and (B) the transformation (frying) process.

Puffing dough: puffing dough in a high-temperature medium (such as baking and frying) is a standard cooking method across cultures. Yet, puffable doughs are usually high-hydration dough that collapses to form a puddle. In this design, by using the Carbopol bath as sacrificial support, we were able to fabricate a 3D flower-shaped dough that expand upon frying (Figure 7).

4 CONCLUSION

In this work, we have adapted embedded printing to the fabrication of freeform and fluidic food materials. Several guidelines were presented for readers to select the appropriate support bath type and print-release method depending on the printing material's properties. Our demos show that printing into a support bath allows us to produce freeform structures with high fidelity and expand the materials library to include fluidic food inks. Beyond enabling personal fabrication of cuisines, our approach will also enable future HCI works to explore and unlock the interaction potential of morphable and instrumentalized edible interfaces.

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